

TrES-5b

R-0972

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_180603 · created 2026-07-29T17:43:03+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458272.8655 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.104 +/- 0.045
Transit depth (Rp/Rs)^2	1.08 +/- 0.94 [%]
Orbital Inclination (inc)	83.06 +/- 2.73
Airmass coefficient 1 (a1)	1.153 +/- 0.029
Airmass coefficient 2 (a2)	-0.111 +/- 0.025
Scatter in the residuals of the lightcurve fit is	2.48 %
Best Comparison Star	None
Optimal Aperture	2.82
Optimal Annulus	6.27
Transit Duration (day)	0.054 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.104 $\pm$ 0.045 vs 0.142 (deviation 0.84 σ)
Duration fit vs published	0.054 d vs 0.0758 d (deviation 0.84 σ)
Mid-time O-C	5.0 min
Depth SNR	2.3
Residual scatter	2.48 %

Light curve

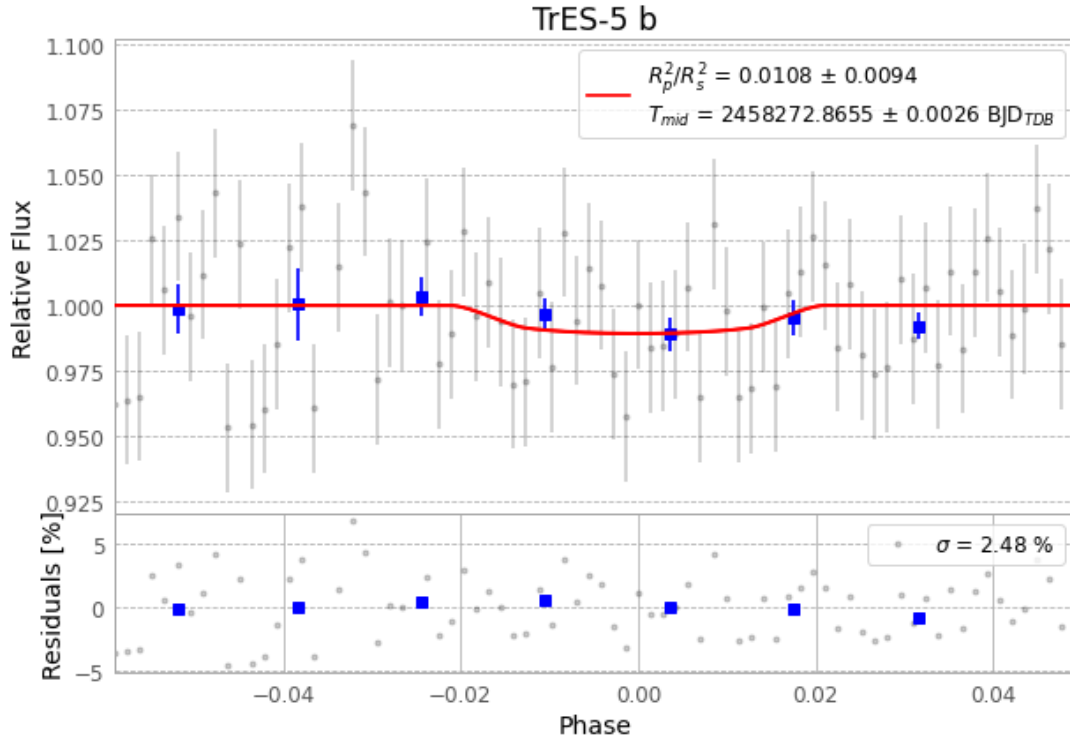


Figure 1 — FinalLightCurve_TrES-5 b_2018-06-02.png (SHA-256 0ebd76fa84c887d3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [454, 236], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 946d386784c60fe2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

78 FITS frames (51 MB), TRES-5180603063912.FITS → TRES-5180603103012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-180603.log (a0317f224623...), FinalLightCurve_TrES-5 b_2018-06-02.png (0ebd76fa84c8...), FinalLightCurve_TrES-5 b_2018-06-02.csv (c33070fd2c4c...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 17:43Z.